

Cambridge International AS & A Level

CANDIDATE
NAME

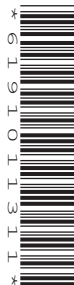
--

CENTRE
NUMBER

--	--	--	--	--

CANDIDATE
NUMBER

--	--	--	--



FURTHER MATHEMATICS

9231/22

Paper 2 Further Pure Mathematics 2

May/June 2021

2 hours

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [].

This document has **16** pages.

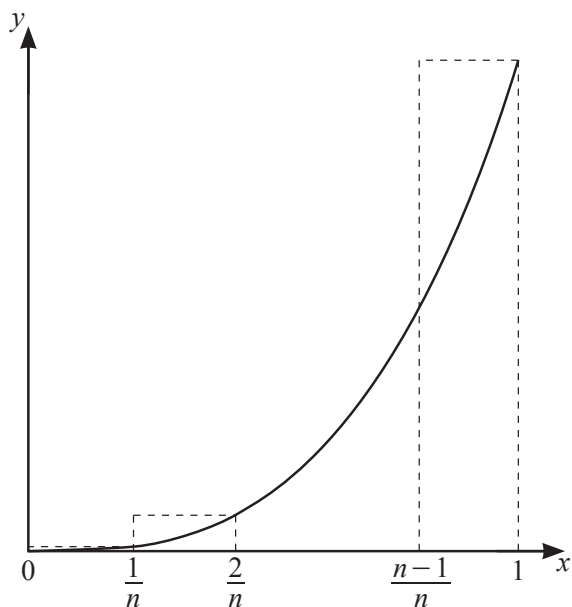
- 2** The variables x and y are related by the differential equation

$$\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = 2x + 1.$$

- (a) Find the general solution for y in terms of x . [6]

[illegible]

- (b) State an approximate solution for large positive values of x . [1]



The diagram shows the curve with equation $y = x^3$ for $0 \leq x \leq 1$, together with a set of n rectangles of width $\frac{1}{n}$.

- (a) By considering the sum of the areas of these rectangles, show that $\int_0^1 x^3 dx < U_n$, where

$$U_n = \left(\frac{n+1}{2n}\right)^2. \quad [4]$$

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins, text, or other markings on the paper.

- (b) Use a similar method to find, in terms of n , a lower bound L_n for $\int_0^1 x^3 dx$. [4]

[illegible]

- (c) Find the least value of n such that $U_n - L_n < 10^{-3}$. [2]

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins, text, or other markings on the paper.

(b) Given that z is an n th root of unity and $z \neq 1$, deduce that $1 + z + z^2 + \dots + z^{n-1} = 0$. [2]

(c) Given instead that $z = \frac{1}{3}(\cos \theta + i \sin \theta)$, use de Moivre's theorem to show that

[illegible]

This image shows a blank sheet of white paper with ten horizontal dashed lines, typical of primary-ruled notebook paper. The lines are evenly spaced and extend across the width of the page. There is no handwriting or other markings on the paper.

- 7 (a) It is given that $y = \operatorname{sech}^{-1}\left(x + \frac{1}{2}\right)$.

Express $\cosh y$ in terms of x and hence show that $\sinh y \frac{dy}{dx} = -\frac{1}{\left(x + \frac{1}{2}\right)^2}$. [3]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

- (b) Find the first three terms in the Maclaurin's series for $\operatorname{sech}^{-1}\left(x + \frac{1}{2}\right)$ in the form

$$\ln a + bx + cx^2,$$

where a , b and c are constants to be determined. [7]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(a) Find $\frac{dx}{dt}$ and show that $\frac{dy}{dt} = 1 - \sinh^2 t$. [3]

[illegible]

(b) (i) Show that $A = \pi \int_0^1 \left(\frac{3}{2}t - \frac{1}{4} \sinh 2t \right) (1 + \cosh 2t) dt$. [4]

This image shows a full page of white paper with horizontal dashed lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

- (ii) Hence find A in terms of π , $\sinh 2$ and $\cosh 2$. [6]

This image shows a full page of a document template designed for handwriting practice or general writing. It consists of approximately 28 evenly spaced, horizontal dotted lines running across the width of the page. The background is plain white, and there are no margins, headers, footers, or other markings present.

[illegible]

Cambridge Assessment International Education is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which itself is a department of the University of Cambridge.